Phase 4: Testing, Validation & Security

System Engineering Workbook & Quality Assurance Sign-Off

1. Objective & Scope

The objective of Phase 4 is to verify that the developed custom application functions as expected, matches all structural functional constraints, and aligns with organizational security and runtime tracking best practices. This lifecycle step ensures the completion of end-to-end automation pipelines, validates layout accuracy across environments, and establishes a verified data framework restricting unauthorized modifications to ticket records.

2. Core Data Validation Criteria

Before deployment verification, the schema definitions are checked against the following automated parameters:

- **Mandatory Fields Capture:** Essential operational parameters must be strictly evaluated to prevent empty string submissions.
- **Auto-Number Engine Tracking:** Sequential record indexing algorithms must evaluate without duplicate collisions.
- **Taxonomy Integration:** Category and dependent Subcategory values must match standardized backend choices.
- **Foreign Key Constraints:** Reference data pills must resolve cleanly to the target User (*sys_user*) and Group (*sys_user_group*) tables.

3. Functional Testing Procedures & Scenarios

Activity 1: Test Scenario 1 (Network Workflow Validation)

This procedure tests the sub-string execution behavior when parsing network-specific terminology.

Step 1. Launch the live table entry form for *Incident WorkFlow*.
Step 2. Select a valid user from the reference lookup for the **Caller** field.
Step 3. Input string exactly into field: **Short Description:** WiFi not working in library
Step 4. Click the form header context control and choose **Save**.
Step 5. Reload the current page session.
Expected Outcome:
- *Category* auto-populates instantly to: **Network**
- *Subcategory* auto-populates instantly to: **Wi-Fi**
- Transactional tracking engine sends out a notification to the caller's target

Email ID.

Activity 2: Transactional System Log Email Inspection

This validation track ensures that transactional notification templates are dispatched cleanly without runtime trace script crashes.

Step 1. Expand the **All** application context dashboard menu panel.
Step 2. Input text string Emails into the navigation filter block.
Step 3. Locate the **System Logs** application node ➔ Click on **Emails** module list.
Step 4. Apply a filtering constraint search parameter: **Subject Contains** Your Request for the issue has been submitted.
Step 5. Open the matching record entry from the system registry table view.
Step 6. Go to the **Related Links** sub-panel cluster and select **Preview Mail**.
Expected Outcome: An HTML render overlay loads cleanly containing accurate, personalized data values matching the tracking ticket context.

Activity 3: Test Scenario 2 (Hardware Workflow Validation)

This procedure tests the multi-branch execution tracking paths when peripheral device hardware flags are encountered.

Step 1. Re-open a fresh blank table entry layout form layout sheet canvas.
Step 2. Establish a valid account map assignment inside the **Caller** lookup context node.
Step 3. Input string exactly into field: **Short Description:** Projector not turning on.
Step 4. Commit form data entries via the header workspace **Save** command action button.
Step 5. Refresh the live layout viewport window frame.
Expected Outcome:
• *Category* auto-populates instantly to: **Hardware**
• *Subcategory* auto-populates instantly to: **Projector**
• Automated email payload triggers accurately and drops clean tracking traces into transmission queues.

4. Test Automation Traceability Matrix

Test ID	Input Conditions	Expected Fields	Notification Status
TS-001	SD: *'WiFi not working in library'*	Cat: **Network** Subcat: **Wi-Fi**	Dispatched & Verified in Logs
TS-002	SD: *'Projector not turning on'*	Cat: **Hardware** Subcat: **Projector**	Dispatched & Verified in Logs